WMG Postgraduate Student-facing Marking Rubric		
Student ID: **XXXXXXXXX**	Component Title and Weighting: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**	Marker: **xxxxxxxxxxxxxxxxx**

Learning Outcome or Evaluative Criteria	**Comments**	**80+** **Outstanding**	**70-79** **Distinction**	**60-69** **Good Pass**	**50-59** **Pass**	**40-49** **Marginal Fail**	**<40** **Fail**
LO1. Critically evaluate and explain the key differences between traditional machine learning and deep learning approaches.	Marker comments	An exemplary answer, showing complete mastery of the learning outcome, with an exceptionally developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are exceeded, and the answer is free from errors. The answer demonstrates originality of thought, with strong critical reflection and the ability to tackle issues not previously encountered. Ideas are explained with great lucidity and in an extremely structured manner. Sophisticated and in-depth comparison of ML and DL paradigms. Demonstrates advanced understanding, uses precise technical language, and presents a highly contextualised justification for the approach taken, well-integrated with the problem context.	An excellent answer, showing mastery of the learning outcome, with a highly developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are covered, and the work is free from all but very minor errors. There is good critical reflection and the ability to tackle issues not previously encountered. Ideas are explained very clearly and in a highly structured manner. Strong, critical comparison that reflects sound theoretical and practical knowledge. Justification for the chosen DL approach is well-integrated with the problem context.	A strong answer, showing a sound grasp of the learning outcome and a skilful attempt at analysis, synthesis and application of concepts, models, and techniques. All requirements of the learning outcome are covered, but there may be some minor errors. There is some critical reflection, and a reasonable attempt is made to tackle issues not previously encountered. Ideas are explained clearly and in a well-organised manner, with some minor exceptions. Offers a clear, accurate, and well-reasoned comparison. Demonstrates good technical understanding and articulates relevant trade-offs linked to the problem.	A satisfactory answer, showing a grasp of the learning outcome, but analysis, synthesis and application of concepts, models and techniques is mechanical, with a heavy reliance on course materials. The requirements of the learning outcome are covered, but with little or no critical reflection and limited ability to tackle issues not previously encountered. Ideas are explained adequately, but with some confusion and a lack of organisation. Provides a basic but valid explanation of ML vs. DL, with some relevant distinctions and generic justification for choosing DL.	An unsatisfactory answer. The learning outcome is not met, but it would not take too much to meet it. There is a weak attempt at analysis, synthesis and application of concepts, models, and techniques. Only some of the requirements of the learning outcome are covered. Inability to reflect critically and difficulty in beginning to address issues not previously encountered. Ideas are poorly explained and organised. Shows limited or partial understanding. Comparison is superficial, with some factual inaccuracies or weak justification for the chosen method. Lack or insufficient references and citations.	An inadequate answer, and there are extremely serious gaps in knowledge and many areas of confusion. Few or none of the requirements of the learning outcome are covered. There is a lack of serious engagement with the learning outcome, and there is no evidence that issues not previously encountered can be addressed. The levels of expression and organisation in the work are very inadequate. Demonstrates little or no understanding of the differences between ML and DL. No meaningful comparison. Terminology and conceptual foundations are incorrect or missing. Lack or insufficient references and citations.

Learning Outcome or Evaluative Criteria	**Comments**	**80+** **Outstanding**	**70-79** **Distinction**	**60-69** **Good Pass**	**50-59** **Pass**	**40-49** **Marginal Fail**	**<40** **Fail**
LO2. Critically evaluate deep learning and artificial intelligence solutions and their implications.	Marker comments	An exemplary answer, showing complete mastery of the learning outcome, with an exceptionally developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are exceeded, and the answer is free from errors. The answer demonstrates originality of thought, with strong critical reflection and the ability to tackle issues not previously encountered. Ideas are explained with great lucidity and in an extremely structured manner. Technically exemplary and professional-level implementation alongside documentation. Evaluation is thorough, insightful, and well-justified. Ethical, ecological, and societal implications are critically analysed with maturity, originality, and contextual depth.	An excellent answer, showing mastery of the learning outcome, with a highly developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are covered, and the work is free from all but very minor errors. There is good critical reflection and the ability to tackle issues not previously encountered. Ideas are explained very clearly and in a highly structured manner. Strong implementation with thoughtful design and use of tools, and well-documented. Evaluation is comprehensive and includes interpretability. The reflection presents a clear, critical discussion of implications.	A strong answer, showing a sound grasp of the learning outcome and a skilful attempt at analysis, synthesis and application of concepts, models, and techniques. All requirements of the learning outcome are covered, but there may be some minor errors. There is some critical reflection, and a reasonable attempt is made to tackle issues not previously encountered. Ideas are explained clearly and in a well-organised manner, with some minor exceptions. Solution is well-structured, reproducible, and reasonably documented. Evaluation of the model is clear and includes interpretability. The written reflection explores ethical, societal, and/or environmental considerations with insight.	A satisfactory answer, showing a grasp of the learning outcome, but analysis, synthesis and application of concepts, models and techniques is mechanical, with a heavy reliance on course materials. The requirements of the learning outcome are covered, but with little or no critical reflection and limited ability to tackle issues not previously encountered. Ideas are explained adequately, but with some confusion and a lack of organisation. Implementation is functional and includes the main pipeline stages. Code is reproducible, though it may lack clarity or detail. Evaluation uses some appropriate metrics. Ethical or contextual implications are acknowledged, though analysis is basic.	An unsatisfactory answer. The learning outcome is not met, but it would not take too much to meet it. There is a weak attempt at analysis, synthesis and application of concepts, models, and techniques. Only some of the requirements of the learning outcome are covered. Inability to reflect critically and difficulty in beginning to address issues not previously encountered. Ideas are poorly explained and organised. The code runs but is poorly structured or barely functional. Key stages (e.g. preprocessing, training, evaluation) are underdeveloped. Evaluation of the solution is vague. Ethical and social implications are only briefly mentioned. Lack or insufficient references and citations.	An inadequate answer and there are extremely serious gaps in knowledge and many areas of confusion. Few or none of the requirements of the learning outcome are covered. There is a lack of serious engagement with the learning outcome, and there is no evidence that issues not previously encountered can be addressed. The levels of expression and organisation in the work are very inadequate. Implementation is missing, incomplete, or fails to address the stated problem. Code lacks structure or documentation. Evaluation is absent or inappropriate. Ethical and contextual considerations are missing or confused. Lack or insufficient references and citations.

Learning Outcome or Evaluative Criteria	Comments	80+ Outstanding	70-79 Distinction	60-69 Good Pass	50-59 Pass	40-49 Marginal Fail	<40 Fail
LO3. Demonstrate an understanding of emerging trends and research development in AI.	Marker comments	An exemplary answer, showing complete mastery of the learning outcome, with an exceptionally developed and mature ability to analyse, synthesise and apply concepts, models, and techniques. All requirements of the learning outcome are exceeded, and the answer is free from errors. The answer demonstrates originality of thought, with strong critical reflection and the ability to tackle issues not previously encountered. Ideas are explained with great lucidity and in an extremely structured manner. Demonstrates outstanding understanding of cutting-edge AI trends and research. Shows originality in proposing how trends could significantly evolve or transform the current solution or field.	An excellent answer, showing mastery of the learning outcome, with a highly developed and mature ability to analyse, synthese and apply concepts, models, and techniques. All requirements of the learning outcome are covered, and the work is free from all but very minor errors. There is good critical reflection and the ability to tackle issues not previously encountered. Ideas are explained very clearly and in a highly structured manner. Strong awareness of current AI research and innovation. Trends are contextualised, with clear links to the problem domain and project future.	A strong answer, showing a sound grasp of the learning outcome and a skilful attempt at analysis, synthesis and application of concepts, models, and techniques. All requirements of the learning outcome are covered, but there may be some minor errors. There is some critical reflection, and a reasonable attempt is made to tackle issues not previously encountered. Ideas are explained clearly and in a well-organised manner, with some minor exceptions. Good understanding of relevant AI developments. Clearly explains one or more trends and how they could shape or enhance future work.	A satisfactory answer, showing a grasp of the learning outcome, but analysis, synthesis and application of concepts, models and techniques is mechanical, with a heavy reliance on course materials. The requirements of the learning outcome are covered, but with little or no critical reflection and limited ability to tackle issues not previously encountered. Ideas are explained adequately, but with some confusion and a lack of organisation. Identifies at least one relevant trend or research direction. Some attempt to relate it to the problem and solution.	An unsatisfactory answer. The learning outcome is not met, but it would not take too much to meet it. There is a weak attempt at analysis, synthesis and application of concepts, models, and techniques. Only some of the requirements of the learning outcome are covered. Inability to reflect critically and difficulty in beginning to address issues not previously encountered. Ideas are poorly explained and organised. Some awareness of AI trends, but the discussion lacks clarity or accuracy. Connection to the chosen problem is weak or unclear. Lack or insufficient references and citations.	An inadequate answer, and there are extremely serious gaps in knowledge and many areas of confusion. Few or none of the requirements of the learning outcome are covered. There is a lack of serious engagement with the learning outcome, and there is no evidence that issues not previously encountered can be addressed. The levels of expression and organisation in the work are very inadequate. No meaningful reference to current AI trends or research. Mentions are vague or incorrect. No link to the project context. Lack or insufficient references and citations.

IF GROUP WORK: Your individual mark may be in a different band than your performance against each learning outcome in this rubric, because of the process of peer adjustment of marks for group work.

What went well? What could be improved and how? ***Marker additional comments***
Student reflections and forward planning. ***Space for students to write reflections on their work and note action plans***